

CS4246 / CS5446

T5 Midterm Review Tutorial

Semester 1, AY2026–27

Concept recaps, tutorial questions and worked solutions

Review topics

Question	Topic	Main task
1	Classical planning	Check preconditions and apply effects
2	Reachable sets	Interpret lower and upper bounds
3	Preferences	Compare lotteries using expected utility
4	Decision theory	Evaluate utility of final wealth
5	Markov decision processes	Represent unfinished actions
6	Passive reinforcement learning	Compare MC, TD(0) and ADP

This tutorial covers only some of the topics relevant to the midterm.

Classical planning recap

A state s is a set of true ground facts, such as $\text{At}(P_1, \text{SFO})$.

An action schema uses variables. A ground action replaces them with objects, such as $\text{Load}(C_2, P_2, \text{JFK})$.

Applicable in s : every precondition is true in s .

$\text{Result}(s, a) = (s \setminus \text{Delete}(a)) \cup \text{Add}(a)$

Facts outside the action's effects remain true.

QUESTION

Q1 setup: air cargo

Initial state

$\text{At}(C_1, \text{SFO}) \wedge \text{At}(C_2, \text{JFK}) \wedge \text{At}(P_1, \text{SFO}) \wedge \text{At}(P_2, \text{JFK})$
 $\wedge \text{Cargo}(C_1) \wedge \text{Cargo}(C_2) \wedge \text{Plane}(P_1) \wedge \text{Plane}(P_2)$
 $\wedge \text{Airport}(\text{JFK}) \wedge \text{Airport}(\text{SFO})$

Goal

$\text{At}(C_1, \text{JFK}) \wedge \text{At}(C_2, \text{SFO})$

C_1, C_2 are cargo. P_1, P_2 are planes. JFK and SFO are airports.

Q1 setup: action schemas

Load(c, p, a)

Pre: $At(c, a) \wedge At(p, a) \wedge Cargo(c) \wedge Plane(p) \wedge Airport(a)$

Effect: $\neg At(c, a) \wedge In(c, p)$

Unload(c, p, a)

Pre: $In(c, p) \wedge At(p, a) \wedge Cargo(c) \wedge Plane(p) \wedge Airport(a)$

Effect: $At(c, a) \wedge \neg In(c, p)$

Fly(p, from, to)

Pre: $At(p, from) \wedge Plane(p) \wedge Airport(from) \wedge Airport(to)$

Effect: $\neg At(p, from) \wedge At(p, to)$

QUESTION

Q1a: applicable Fly actions

What are all applicable concrete instances of $\text{Fly}(p, \text{from}, \text{to})$ in the state below, given the supplied action schemas?

$\text{At}(P_1, \text{JFK}) \wedge \text{At}(P_2, \text{SFO}) \wedge \text{Plane}(P_1) \wedge \text{Plane}(P_2) \wedge \text{Airport}(\text{JFK}) \wedge \text{Airport}(\text{SFO})$

$\text{Fly}(p, \text{from}, \text{to})$

Pre: $\text{At}(p, \text{from}) \wedge \text{Plane}(p) \wedge \text{Airport}(\text{from}) \wedge \text{Airport}(\text{to})$

Effect: $\neg \text{At}(p, \text{from}) \wedge \text{At}(p, \text{to})$

This part places P_1 at JFK and P_2 at SFO.

Q1a solution: four ground actions

The current location fixes from. Either airport can be to.

Plane location	Applicable instances
P_1 at JFK	
P_2 at SFO	

The schema has no from $\neq$ to precondition.

Flights to the same airport are allowable, though not useful.

QUESTION

Q1b: applying Load

What is the result of executing $\text{Load}(C_2, P_2, \text{JFK})$ from the initial state?

Init: $\text{At}(C_1, \text{SFO}) \wedge \text{At}(C_2, \text{JFK}) \wedge \text{At}(P_1, \text{SFO}) \wedge \text{At}(P_2, \text{JFK})$
 $\wedge \text{Cargo}(C_1) \wedge \text{Cargo}(C_2) \wedge \text{Plane}(P_1) \wedge \text{Plane}(P_2)$
 $\wedge \text{Airport}(\text{JFK}) \wedge \text{Airport}(\text{SFO})$

$\text{Load}(c, p, a)$

Pre: $\text{At}(c, a) \wedge \text{At}(p, a) \wedge \text{Cargo}(c) \wedge \text{Plane}(p) \wedge \text{Airport}(a)$

Effect: $\neg \text{At}(c, a) \wedge \text{In}(c, p)$

Q1b solution: the resulting state

Load(C_2 , P_2 , JFK) is applicable: all five preconditions hold.

Delete - Add -

$At(C_1, SFO) \wedge At(P_1, SFO) \wedge At(P_2, JFK)$
 $\wedge Cargo(C_1) \wedge Cargo(C_2) \wedge Plane(P_1) \wedge Plane(P_2)$
 $\wedge Airport(JFK) \wedge Airport(SFO) \wedge In(C_2, P_2)$

All other initial-state facts remain unchanged.

Reachable sets recap

Fix a high-level plan and its start state. Let R contain the endpoints reachable by its executable primitive refinements.

$$R^- \subseteq R \subseteq R^+$$

R^- : pessimistic set. Each included endpoint has a refinement.

R^+ : optimistic set. Every actually reachable endpoint lies inside.

For goal set G : $R^+ \cap G = \emptyset$ rules out a successful refinement.

$R^- \cap G \neq \emptyset$ proves that a successful refinement exists.

QUESTION

Q2: what can the bounds tell us?

Suppose the optimistic reachable set of a high-level plan is a superset of the goal set.

$$G \subseteq R^+$$

Can we conclude anything about whether the plan will achieve the goal?

What if the pessimistic reachable set does not intersect with the goal set: $R^- \cap G = \emptyset$? Explain.

Q2 solution:---

Example: $G = \{g\}$, $R^- = \{b\}$, $R^+ = \{b, g\}$.

Both $G \subseteq R^+$ and $R^- \cap G = \emptyset$ hold.

Possible actual set R	Successful refinement?
$\{b\}$	No goal endpoint is reachable
$\{b, g\}$	Yes, a refinement reaches g

R^+ may include unreachable goals.

R^- may omit reachable goals.

Expected utility recap

Assume preferences have an expected-utility representation.
Let a random payoff X equal x_i with probability p_i .

$$EU(X) = \sum_i p_i U(x_i)$$

X is strictly preferred to Y exactly when $EU(X) > EU(Y)$.

A certain payoff c has expected utility $U(c)$.

Equal expected money can still give different expected utility.

QUESTION

Q3: Alex's two games

Game 1: flip a fair coin once. Heads pays \$100.
Tails pays nothing.

Game 2: flip a fair coin twice. Each head pays \$50.
Each tail pays nothing.

Alex prefers Game 2 to Game 1. Argue that Alex would prefer receiving \$50 directly to participating in Game 1.

Assume independent fair flips and expected-utility preferences.

Q3 working: payoff distributions

Total payoff	Game 1 probability	Game 2 probability
\$0	1/2	1/4 (TT)
\$50	0	1/2 (HT or TH)
\$100	1/2	1/4 (HH)

$$EU(\text{Game 1}) = 0.5U(\$0) + 0.5U(\$100)$$

$$EU(\text{Game 2}) = 0.25U(\$0) + 0.5U(\$50) + 0.25U(\$100)$$

Q3 solution: the utility inequality

Alex's preference gives $EU(\text{Game 1}) < EU(\text{Game 2})$:

$EU(\text{Game 1}) =$

Therefore, Alex strictly prefers a certain \$50 to Game 1.

Expected utility and final wealth

Let W be current wealth and X the random gain or loss.

Final wealth = $W + X$

$$EU = E[U(W + X)] = \sum p_i U(W + x_i)$$

Here $U(x) = \ln(x)$, so every possible final wealth must be positive.

$$U'(x) = 1/x > 0 \quad U''(x) = -1/x^2 < 0$$

Increasing utility rewards wealth. Concavity expresses risk aversion.

QUESTION

Q4: Alice's two investments

Alice has utility $U(x) = \ln(x)$, where x is total wealth.

Alternative	Outcome 1	Outcome 2
A	Win \$10,000 with $p = 0.2$	Win \$1,000 with $p = 0.8$
B	Win \$3,000 with $p = 0.9$	Lose \$2,000 with $p = 0.1$

(a) $W = \$2,500$ (b) $W = \$5,000$ (c) $W = \$10,000$

For each wealth level, should she choose A or B?

(d) Is this pattern of choices reasonable? Why or why not?

SOLUTION

Q4(a–c): Expected-utility calculations

$$EU(A; W) = 0.2 \ln(W + 10000) + 0.8 \ln(W + 1000)$$

$$EU(B; W) = 0.9 \ln(W + 3000) + 0.1 \ln(W - 2000)$$

Current wealth W	EU(A; W)	EU(B; W)	Choice
\$2,500			
\$5,000			
\$10,000			

The gap at \$10,000 is only 0.001064. Keep enough precision.

Q4(d): Interpreting the A–B–A pattern

Alice chooses A, then B, then A as wealth increases

Log utility is risk averse, but it does not require at most one preference switch as wealth changes.

Markov state and the Bellman equation

A state must contain the information needed to predict what happens next.

$$P(S_{t+1} \mid S_0, A_0, \dots, S_t, A_t) = P(S_{t+1} \mid S_t, A_t)$$

An MDP specifies states S , legal actions $A(s)$, transitions P , and rewards R . The return objective also needs a horizon or discount γ .

$$V^*(s) = \max_a \sum P(s' \mid s, a) [R(s, a, s') + \gamma V^*(s')]$$

The sum is over next states s' . The maximum is over legal actions at s .

An action still in progress can change both of these.

QUESTION

Q5: Robot navigation with delayed movements

A robot moves in a 10×10 grid. Cell (10,10) is the goal.

Each time step in a non-goal cell costs 0.05. Reaching the goal gives +100 and ends the episode.

The robot chooses UP, DOWN, LEFT or RIGHT.

Direction of movement	Probability
Intended direction	0.8
First orthogonal direction	0.1
Second orthogonal direction	0.1

QUESTION

Q5: Action duration and the modelling task

An action starts at clock time t .

Probability	What happens
0.5	Movement finishes at $t + 1$.
0.5	At $t + 1$ the robot stays in its cell. Movement finishes at $t + 2$.

A new movement action is allowed only after the old one finishes.

Specify the state space, legal actions, transition probabilities and reward function of an MDP, including unfinished actions.

Optional: determine the optimal policy at $t = 2$ and name the method.

Q5: State space and legal actions

Write $x = (i,j)$, with $i,j \in \{1,\dots,10\}$, and $g = (10,10)$.

State	Meaning	Legal actions
??	Ready at position x	UP, DOWN, LEFT, RIGHT
??	Action a is pending. One step remains.	??
$(g,0)$	Terminal goal	Episode ends

The state records the pending action a , as well as the delay.

A delay flag alone cannot distinguish a pending UP from a pending RIGHT.

Q5: Spatial and temporal transitions

Let $T(x' \mid x, a)$ be the spatial transition model: 0.8 intended, 0.1 for each orthogonal direction. Off-grid outcomes stay at x .

Assume movement direction and action duration are independent.

Ready, movement finishes in one step

$$P((x', 0) \mid (x, 0), a) =$$

Ready, movement is delayed

$$P((x, a, 1) \mid (x, 0), a) =$$

Pending, movement finishes after WAIT

$$P((x', 0) \mid (x, a, 1), \text{WAIT}) =$$

Q5: Rewards and a transition example

Use one MDP transition for each elapsed clock step.

For a nonterminal state: $R(s,a,s') =$

The indicator equals 1 on goal entry and 0 otherwise.

A delayed step also costs 0.05. After termination, future reward is 0.

Example: at ready position (5,5), select UP.

Next augmented state	Probability	Reward
Ready (5,6), (4,5), (6,5)	0.40, 0.05, 0.05	-0.05 each
Pending ((5,5),UP,1)	0.50	-0.05

From the pending state, WAIT completes UP with probabilities 0.8 / 0.1 / 0.1.

Policy evaluation and learning targets

A fixed policy π chooses the actions. Passive RL estimates its value.

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots$$

$$U_{\pi}(s) = E_{\pi}[G_t \mid S_t = s]$$

$U(s)$ denotes our current estimate of the policy value $U_{\pi}(s)$.

$$U(s) \leftarrow U(s) + \alpha[\text{target} - U(s)]$$

Method	Target
Monte Carlo	A sampled return G_t
TD(0)	$r + \gamma U(s')$ from one sampled transition
ADP	An expected Bellman backup using a model

QUESTION

Q6: Learning values under a fixed policy

The policy always takes S_0 (start) to S_1 (intermediate), then S_1 to S_2 (terminal). The episode ends at S_2 .

$$R(S_0 \rightarrow S_1) = 0 \qquad R(S_1 \rightarrow S_2) = 1$$

We want to learn $U\pi(s)$ for $s \in \{S_0, S_1, S_2\}$.

- (a) MC: When does each state update? What data must it wait for?
- (b) TD(0): When and how does it update? Why is it bootstrapping?
- (c) ADP: What extra information does it need? How does it solve or iteratively refine the value function?

SOLUTION

Q6(a): Monte Carlo uses completed returns

MC waits until the episode ends, then computes each visited state's return.

$$U(s) \leftarrow U(s) + \alpha[G_t - U(s)]$$

For the observed episode $S_0 \rightarrow S_1 \rightarrow S_2$:

G at S_0 = ??

G at S_1 = ??

If U starts at 0 and $\alpha = 1$, one episode gives

$U(S_0) = ??$

$U(S_1) = ??$

$U(S_2) = ??$

The reward +1 arrives on entry to S_2 . Its remaining return is zero.

Q6(b): TD(0) updates after each transition

$$U(s) \leftarrow U(s) + \alpha[r + \gamma U(s') - U(s)]$$

Example: initial values 0, $\alpha = 1$, online updates in observed order.

Observed transition	Target at that moment	Updated value
$S_0 \rightarrow S_1$, reward 0	??	$U(S_0) = ??$
$S_1 \rightarrow S_2$, reward 1	??	$U(S_1) = ??$
Next episode: $S_0 \rightarrow S_1$, $r = 0$	??	$U(S_0) = ??$

TD bootstraps: the target uses the current estimate $U(s')$.

SOLUTION

Q6(c): ADP learns a model and evaluates π

ADP estimates transition probabilities P and expected rewards R from experience, or uses a model if one is available.

$$U(s) = \sum P(s' \mid s, \pi(s)) [R(s, \pi(s), s') + \gamma U(s')]$$

The sum is over s' . Solve these equations or apply repeated backups.

For this deterministic chain, after the model is known:

$$U(S_2) = ? \quad U(S_1) = ?$$

$$U(S_0) = 0 + \gamma U(S_1) = ??$$

The policy is fixed, so this Bellman equation has no maximization.

Q6: Comparing MC, TD(0) and ADP

Feature	MC	TD(0)	ADP
Uses a model?	No	No	Yes, learned or given
Typical update time	Episode end	Each transition	During model-based planning
What the target uses	Actual sampled return	Reward + estimated next-state value	Expected reward + expected next-state value
Bootstraps?	No	Yes	Yes for iterative Bellman backups

All three evaluate the same fixed policy.

Their differences are the information used, update timing and planning.

Midterm review checklist

- Q1 Ground the action, check every precondition, then apply delete/add effects.
- Q2 Use $R^- \subseteq R \subseteq R^+$. Keep existence and impossibility claims distinct.
- Q3 Reduce each lottery to outcomes and probabilities before comparing utility.
- Q4 Use final wealth in U , and keep enough numerical precision.
- Q5 Include pending-action information and charge each elapsed step.
- Q6 Identify the update target, when it becomes available and whether π is fixed.